1. Identity

Document	STAR-PKG-STAR-CDM-EN · v2.2 · 2026-11-02
Package	star-cdm
Version	0.2.2
Capability	star.ssa.cdm
Maturity	Beta (VERIFIED)
Python	⩾3.9
Source commit	dc09f61
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_cdm

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_cdm-0.2.2-py3-none-any.whl
```

FILE	SHA-256
star_cdm-0.2.2-py3-none-any.whl	6f630d78c65bc00ae004d7d8f0b7f0843ebcbcf74ade7e18cdbc2c3899ac2ef3
star_cdm.py	101363f1c48f3a735efa1e1ef3670b58eb75ad1d698b174bd3213b0cc8a5b8a2
star_cdm-0.2.2-source.zip	b4ba1867ccea21347a123ea29c8673f4d62dc32f8d4c1491c72bd78f895b701d

5. What it does, and its limits

Strict parser for CCSDS 508.0-B-1 Conjunction Data Messages in KVN. Standard library only.

XML CDM, CDM 2.0 (508.0-B-2 draft/updates), collision probability computation, any operational use.

- `parse_cdm(text)` → `{"header", "object1", "object2", "warnings"}`: dates as datetimes (CCSDS 502.0 formats only), numbers as floats, units checked against the standard, exactly OBJECT1 then OBJECT2, all obligatory keywords (the set of the standard's example 3.6.2).
- Every malformation raises `CdmFormatError` (unknown keyword, missing obligatory keyword, wrong unit, duplicate, non-numeric, malformed date, wrong object count). Typographic minus signs (U+2212) are normalised and REPORTED.
- `covariance_rtn(obj)` → symmetric 6×6; `consistency_report(cdm)` compares MISS_DISTANCE with the state vectors and RELATIVE_POSITION_R/T/N and flags inconsistencies without editing values.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_cdm_0_2_2,
  author  = {Cavazzini, Andrea},
  title   = {{star-cdm}: Strict CCSDS 508.0-B-1 Conjunction Data Message (KVN) parser with
consistency report},
  version = {0.2.2},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_cdm/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit dc09f61}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

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